

5.3 Cybersecurity	5.3.1 understand the threat to digital systems posed by malware (viruses, worms, Trojans, ransomware, key loggers) and how hackers exploit technical vulnerabilities (unpatched software, out-of-date anti-malware) and use social engineering to carry out cyberattacks
	5.3.2 understand methods of protecting digital systems and data (anti-malware, encryption, acceptable use policies, backup and recovery procedures)

Assessment information

- First assessment: May/June 2022.
- The paper is 1 hour and 30 minutes.
- The paper consists of five questions (one question per topic).
- The paper is out of 75 marks.
- Students must answer all questions.
- The paper consists of multiple-choice, short-, medium- and extended-open-response, tabular and diagrammatic items.
- The paper will include questions that target mathematics at Level 2.
- The paper will include questions that target computer-related mathematics.
- Calculators must not be used in the examination.

Sample assessment materials

A sample question paper and mark scheme for this paper can be found in the *Pearson Edexcel Level 1/Level 2 GCSE (9–1) in Computer Science Sample Assessment Materials (SAMs)* document.

The SAMs will be available on the [GCSE in Computer Science section of our website](#). Please note that the order of the topics will vary in each live assessment. A full list of command words that will be used in assessments are shown in *Appendix 1: Command word taxonomy*.

Paper 2: Application of Computational Thinking

Overview

Learning to program is a core component of a computer science course. Students should be competent at designing, reading, writing and debugging programs. They must be able to apply their skills to solve real problems and produce readable, robust programs.

Content